

Intermediate Quantitative Methods

What is a regression for?

Kevin Munger

Week 2, part 1

European University Institute, SPS

Today

- One regression, three readings

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- One regression, three readings
- Description

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- Description
- Prediction

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- Causal inference

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- Causal inference
- How the three get confused

One regression, three readings

From last week

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- Estimator: the rule we apply to the data

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- Estimator: the rule we apply to the data
- Last week the estimator was a sample mean
- This week it is a regression coefficient
- The harder question: which estimand is it an estimate of?

A regression

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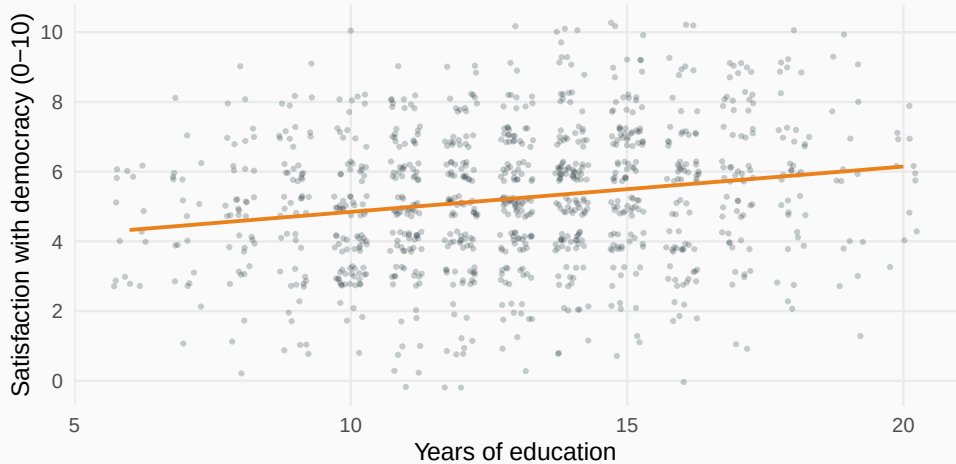
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- β : the slope, how y differs on average per unit of x
- ε : the error, everything else that moves y

Simulated survey, $n = 1,000$



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- Causal: one more year of education would raise a person's satisfaction by 0.13
- Same number, same software output, three different claims

Three different jobs

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- Description, prediction, causal inference

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- Each fails in a different way
- Most confusion in applied work comes from mixing them

Description

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- “Differ by”, not “effect of”

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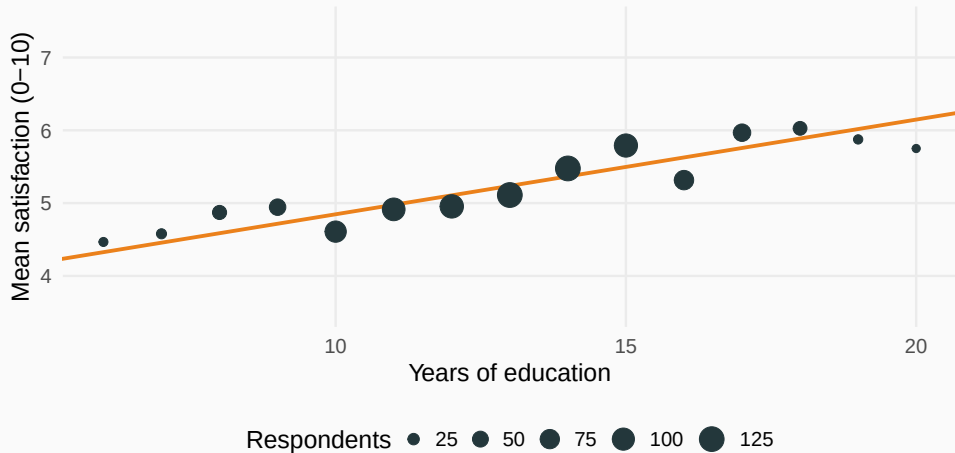
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- A regression line is a summary of those averages
- Binary x : β is exactly the difference between two group means
- Continuous x : the line smooths over many group means

Mean satisfaction at each year of education



What good description needs

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- A defined population
- A sample that represents it
- Measures that capture the concepts
- A summary that fits the pattern: is a line a fair description?
- No assumption about why x and y move together

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- Gerring (2012): description treated as a lesser task
- Causal identification as the price of admission to top journals
- Descriptive findings contorted into causal framing to get published
- Result: a lot of weak causal claims, little careful description

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- Munger, Guess & Hargittai (2021)

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- The contribution in each is a number nobody had

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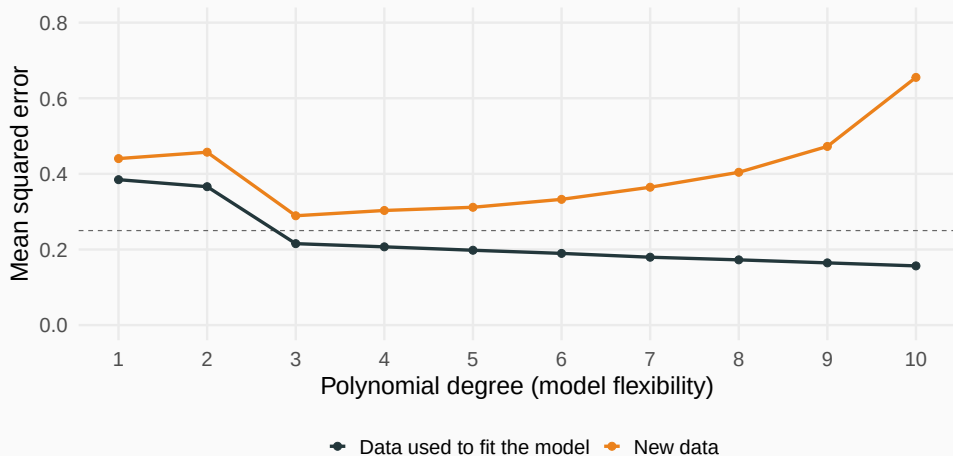
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- “ $\hat{y}$ problems” and “ $\hat{\beta}$ problems” (Mullainathan & Spiess 2017)

Error by model flexibility, 500 simulated datasets of 30



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- Week 8: machine learning

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- Umbrella sales predict rain
- A model can predict well with every coefficient causally meaningless
- A model can identify a real effect and predict badly

Causal inference

A causal claim

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A causal claim

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- A counterfactual: never observed for any one person
- β is causal only if nothing else that drives y differs along with x
- That condition comes from design and knowledge, not from the regression

A confounder

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- Simulation: x has no effect on y at all

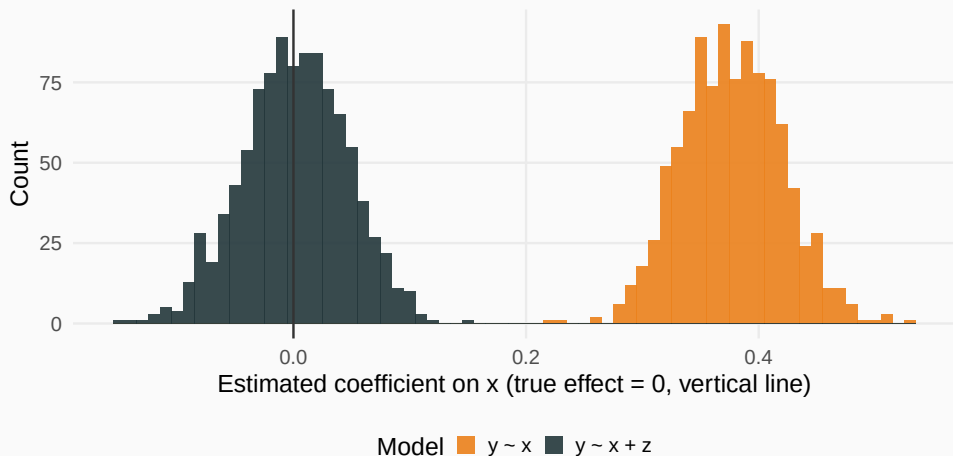
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- Simulation: x has no effect on y at all
- 1,000 datasets, two regressions on each
- $y \sim x$, then $y \sim x + z$

Estimates of an effect that is zero, 1,000 simulated datasets



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- With z : centered on zero
- More data makes the wrong answer more precise, not more correct
- The uncontrolled model still describes and predicts correctly
- Only the causal reading is wrong

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- Week 3: estimands and DAGs

How the three get confused

The Table 2 fallacy

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Causal claims in descriptive clothing

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The wrong yardstick

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- Coefficient signs to judge a prediction model

The wrong yardstick

- R^2 to judge a causal estimate
- Significance stars to judge a description
- Coefficient signs to judge a prediction model
- Our survey regression: $R^2 = 0.04$, and a perfectly good description

Three purposes, three standards

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Target	Needs	Typical failure
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Prediction	$\hat{y}$ for new cases	Fit on unseen data	Overfitting
Causal inference	y if x changed	Design, assumptions	Confounding

- Say which of the three you are doing, early

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- Use the matching standard, and only that one

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- Description usually has to come first

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- Last exercise in Lab 2: is your interpretation descriptive, predictive or causal?

- Part 2: regression anatomy

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- What least squares does

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- What “holding constant” means

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- What least squares does
- What “holding constant” means
- Which assumptions matter, and what happens when they break